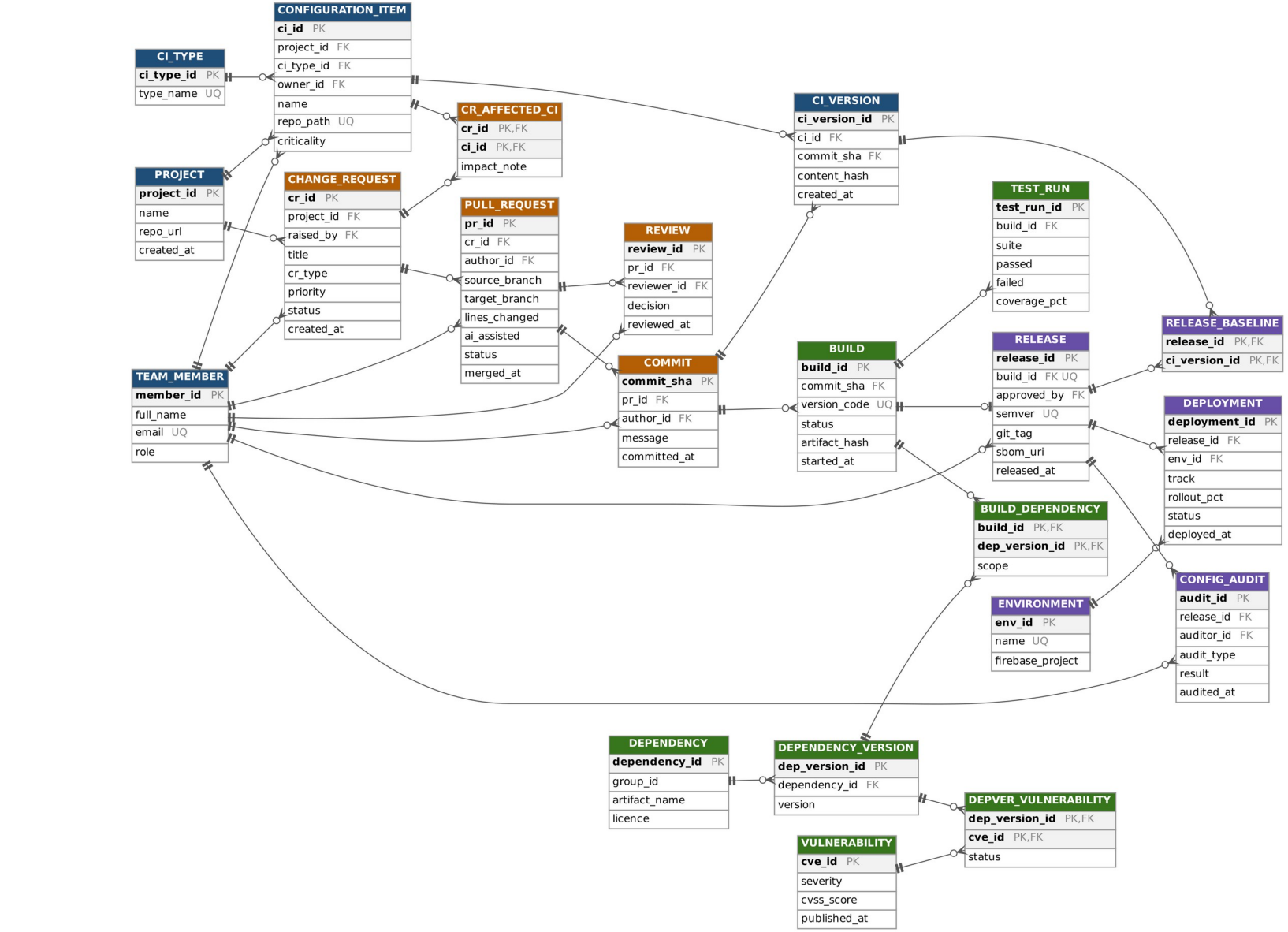
MediAssist SCM / Artifact-Tracking Database — Entity-Relationship Diagram

Name: Navy Lokai **Admission No.:** 672415

USIU-Africa | November 2026 | Repository: https://github.com/NavyLokai/mediassist-scm



Identification · Change control · Build & supply chain · Release & audit Crow's-foot notation: ‖ = exactly one, o< = zero or many, ‖o = zero or one. PK primary key · FK foreign key · UQ unique.

MediAssist SCM / Artifact-Tracking Database — Design Rationale

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1. Purpose and scope

The database is a design-only prototype that records every configuration artifact of MediAssist (Android client, Firebase backend, caregiver dashboard) from the moment a change is requested until the build carrying it is running on users' phones. It turns the thirteen SCM requirements from Part 1 (SCM-01 to SCM-13) into stored, queryable facts, so the team can prove what changed, who approved it, what it was built with, and where it is deployed.

2. Completeness of entities (22 entities in four groups)

- **Identification (SCM-01, 02):** PROJECT, TEAM_MEMBER, CI_TYPE, CONFIGURATION_ITEM (source file, layout, Gradle script, Firebase rule, schema migration, test suite, design asset, document) and CI_VERSION, one row per committed version of an item with its content hash.
- **Change control (SCM-03 to 05):** CHANGE_REQUEST, CR_AFFECTED_CI (impact analysis), PULL_REQUEST (with lines_changed and ai_assisted flags), REVIEW and COMMIT (keyed by the immutable Git SHA).
- **Build and supply chain (SCM-06, 07):** BUILD, TEST_RUN, DEPENDENCY, DEPENDENCY_VERSION, BUILD_DEPENDENCY (the per-build bill of materials), VULNERABILITY (CVE) and DEPVER_VULNERABILITY.
- **Release, deployment and audit (SCM-09 to 12):** RELEASE (SemVer, Git tag, SBOM), RELEASE_BASELINE (the exact CI versions frozen in a release), ENVIRONMENT (dev, staging, production), DEPLOYMENT (track and rollout %) and CONFIG_AUDIT (functional or physical audit).

3. Traceability

The relationships form one unbroken chain: **CHANGE_REQUEST → PULL_REQUEST → COMMIT → CI_VERSION / BUILD → TEST_RUN → RELEASE → RELEASE_BASELINE → DEPLOYMENT → CONFIG_AUDIT**. Because every link is a mandatory foreign key, the chain can be walked in both directions:

- **Forward:** for change request CR-42, list the PRs, reviewers, commits, builds and tests, and the environments where it is live.
- **Backward:** from the versionCode on a caregiver's phone, find the BUILD, its COMMIT, the PR, its approvals and the original request.
- **Impact:** when a new CVE is published, follow DEPVER_VULNERABILITY → BUILD_DEPENDENCY → BUILD → RELEASE → DEPLOYMENT to list every production release that ships the vulnerable library.

RELEASE_BASELINE records configuration identity at release time, so an audit can compare the shipped build against the exact tagged item versions.

4. Normalization (3NF / BCNF)

- **1NF:** all attributes are atomic; there are no lists in columns (reviewers, dependencies and affected items are separate rows).
- **2NF:** the four associative entities (CR_AFFECTED_CI, BUILD_DEPENDENCY, DEPVER_VULNERABILITY, RELEASE_BASELINE) resolve the many-to-many relationships with composite keys, and they hold only attributes of the whole key (e.g. scope, impact_note).
- **3NF:** no transitive dependencies. Licence depends on the library, not the version, so DEPENDENCY is split from DEPENDENCY_VERSION. CI_TYPE and ENVIRONMENT are lookup tables, so names are not repeated. Member details live only in TEAM_MEMBER and are referenced by FK.
- **No derived data:** pass rates, open-CVE counts and approval status are computed by queries, not stored, so they can never contradict the source rows.

5. Key design decisions and integrity rules

- Natural keys where they are immutable (commit_sha, cve_id); surrogate keys elsewhere. BUILD to RELEASE is one-to-zero-or-one (build_id is unique in RELEASE) because not every build is released.
- History tables (COMMIT, CI_VERSION, REVIEW, RELEASE, CONFIG_AUDIT) are append-only: rows are never updated or deleted, which preserves the audit trail.
- Business rules enforced by constraints or triggers: a PR cannot merge without an approving REVIEW (two for critical items); a RELEASE cannot be approved while a linked critical CVE is open; a production DEPLOYMENT requires a passed CONFIG_AUDIT.